WORKED SOLUTION

STEP III 1997, Question 6

Pure • Chebyshev Differential Equation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q6

The key idea

The factor $1-x^2$ next to $x=\cos\theta$ suggests one thing: $1-\cos^2\theta=\sin^2\theta$. Once we change variable carefully with the chain rule, the awkward equation in x becomes the equation for simple harmonic motion in θ. We solve that and use the two side conditions to fix the constants. The recurrence is then a compound-angle identity in disguise. Throughout, n is a non-negative integer. The condition $y_n(x)=(-1)^n y_n(-x)$ only makes sense in that case.

Solution

Changing the variable

Let $x=\cos\theta$ with $0<\theta<\pi$, so $\sin\theta>0$ and $\mathrm{d}x/\mathrm{d}\theta=-\sin\theta\neq 0$. We write y for y_n to keep the notation light. By the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta}=-\frac{1}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}.$$

To differentiate again with respect to x, we differentiate with respect to θ and multiply by $\mathrm{d}\theta/\mathrm{d}x=-1/\sin\theta$:

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2}&=-\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(-\frac{1}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)=\frac{1}{\sin\theta}\left(\frac{1}{\sin\theta}\frac{\mathrm{d}^2y}{\mathrm{d}\theta^2}-\frac{\cos\theta}{\sin^2\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)\\&=\frac{1}{\sin^2\theta}\frac{\mathrm{d}^2y}{\mathrm{d}\theta^2}-\frac{\cos\theta}{\sin^3\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}.\end{aligned}$$

Now substitute, using $1-x^2=\sin^2\theta$ and $x=\cos\theta$:

$$\begin{aligned}(1-x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}-x\frac{\mathrm{d}y}{\mathrm{d}x}+n^2y&=\frac{\mathrm{d}^2y}{\mathrm{d}\theta^2}-\frac{\cos\theta}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}+\frac{\cos\theta}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}+n^2y\\&=\frac{\mathrm{d}^2y}{\mathrm{d}\theta^2}+n^2y.\end{aligned}$$

The two first-derivative terms cancel exactly. So the given equation is equivalent to $\dfrac{\mathrm{d}^2 y_n}{\mathrm{d}\theta^2} + n^2 y_n = 0$, as required.

Ask yourself Why isn't $\dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2}$ simply $\dfrac{1}{\sin^2\theta}\dfrac{\mathrm{d}^2 y}{\mathrm{d}\theta^2}$?

Because $\mathrm{d}y/\mathrm{d}x$ is a *product*: $-1/\sin\theta$ times $\mathrm{d}y/\mathrm{d}\theta$. Both factors depend on θ, so the product rule gives two terms. The extra term is exactly what cancels the $-x\,\mathrm{d}y/\mathrm{d}x$ term, which is why the question has that term in it.

Solving for y_n

For $n \geq 1$ the general solution of $y'' + n^2 y = 0$ is

$$y_n = A\cos n\theta + B\sin n\theta.$$

For $n = 0$ the equation is $y'' = 0$, so $y_0 = A + B\theta$.

The condition at $x = 1$. $x = 1$ corresponds to $\theta = 0$, and $y_n(1) = 1$ gives $A = 1$ in both cases.

The symmetry condition. Replacing x by $-x$ means replacing θ by $\pi - \theta$, since $\cos(\pi - \theta) = -\cos\theta$, and $\pi - \theta$ is again in $[0, \pi]$. For an integer $n \geq 1$,

$$\cos n(\pi - \theta) = (-1)^n \cos n\theta, \qquad \sin n(\pi - \theta) = -(-1)^n \sin n\theta.$$

So $(-1)^n y_n(-x) = (-1)^n[A(-1)^n \cos n\theta - B(-1)^n \sin n\theta] = A\cos n\theta - B\sin n\theta$. Setting this equal to $y_n(x) = A\cos n\theta + B\sin n\theta$ for all θ gives $2B\sin n\theta = 0$, so $B = 0$. For $n = 0$ the condition reads $1 + B\theta = 1 + B(\pi - \theta)$ for all θ, which again forces $B = 0$.

Hence, for every non-negative integer n,

$$\boxed{y_n = \cos n\theta}, \qquad \text{that is,} \quad y_n(x) = \cos(n \arccos x) \quad (|x| \leq 1).$$

The relation $\theta = \arccos x$ takes $[-1, 1]$ onto $[0, \pi]$, which is why the result holds for $|x| \leq 1$. The endpoints $x = \pm 1$ ($\theta = 0, \pi$) are covered by continuity.

The deduction

With $n = 0$ and $n = 1$: $y_0 = \cos 0 = 1$ and $y_1 = \cos\theta = x$.

For the recurrence, add the compound-angle formulae

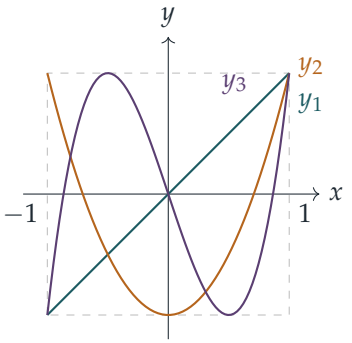
$$\cos(n+1)\theta = \cos n\theta\cos\theta - \sin n\theta\sin\theta, \qquad \cos(n-1)\theta = \cos n\theta\cos\theta + \sin n\theta\sin\theta.$$

The sine terms cancel, leaving $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$. In terms of x,

$$y_{n+1} - 2xy_n + y_{n-1} = 0 \qquad (n \geq 1,\ |x| \leq 1),$$

which is the required result.

Starting from $y_0 = 1$ and $y_1 = x$, the recurrence produces polynomials: $y_2 = 2x^2 - 1$, $y_3 = 4x^3 - 3x$, $y_4 = 8x^4 - 8x^2 + 1$, and so on.



Each y_n stays between -1 and 1 on $[-1, 1]$, as it must, being a cosine.

Checks

- $y_2 = 2x^2 - 1$: then $(1 - x^2) \cdot 4 - x \cdot 4x + 4(2x^2 - 1) = 0$. It satisfies the equation with $n = 2$, and $y_2(1) = 1$, and it is even.
- The parity of y_n matches $(-1)^n$. For example, $y_3 = 4x^3 - 3x$ is odd.
- Numerically, for $x = 0.3$: $\cos(2 \arccos 0.3) = -0.82 = 2(0.3)^2 - 1$. A finite-difference check of the original equation for $y_n = \cos(n \arccos x)$, $0 \leq n \leq 8$, gave residuals below 10^{-5}.

Common slips

- Forgetting the product rule in the second derivative, so the first-derivative terms don't cancel.
- Keeping the $\sin n\theta$ solution. It satisfies the differential equation and $y_n(1) = 1$ (since $\sin 0 = 0$), but it fails the symmetry condition. That condition is there precisely to remove it.
- Forgetting the separate case $n = 0$, where the general solution is $A + B\theta$, not a pair of cosines and sines.

Beyond STEP

The polynomials y_n are the *Chebyshev polynomials* $T_n(x)$. The given equation is Chebyshev's equation. The second solution we discarded, $\sin(n \arccos x)$, is $\sqrt{1 - x^2}$ times a polynomial. Chebyshev polynomials are central to approximation theory. Of all monic polynomials of degree n, $2^{1-n} T_n$ has the smallest maximum size on $[-1, 1]$, which is why the zeros of T_n are the best interpolation points for many numerical methods. For series solutions of equations like this one, and the special functions that come out of them, see Boyce and DiPrima, *Elementary Differential Equations*.

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